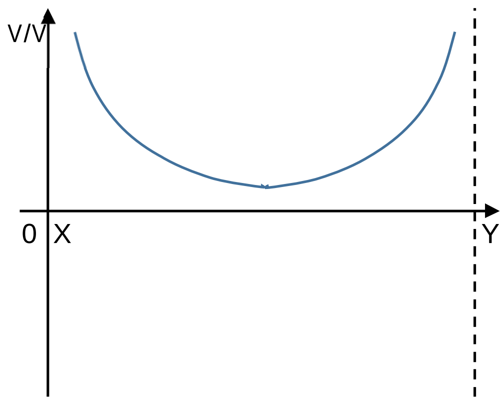
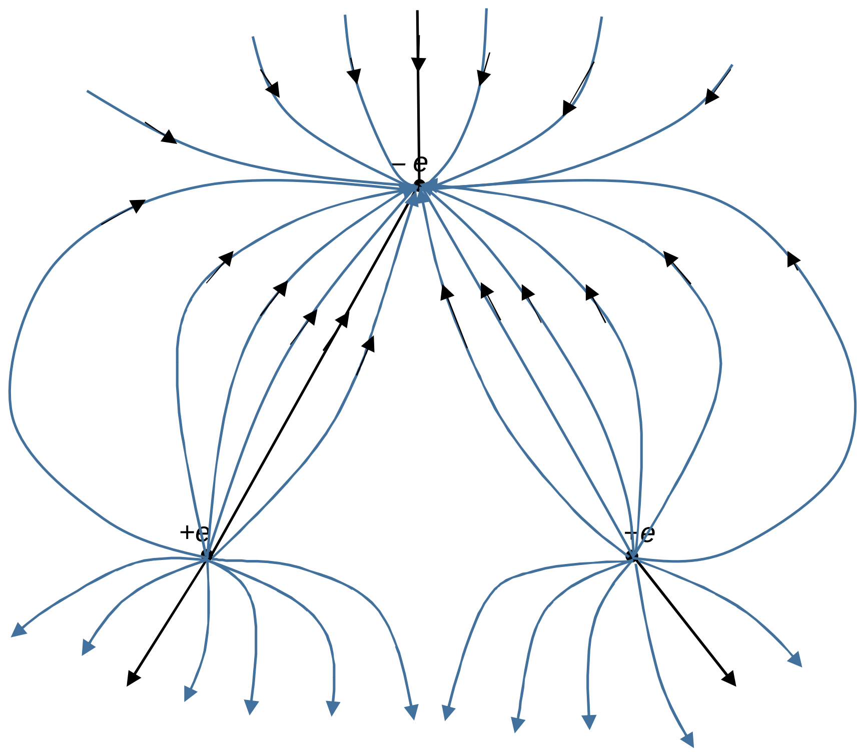
Question		Marks
3(a)	<u>the work done per unit positive charge</u> <u>by an external agent</u> in bringing a small test charge from <u>infinity to that point</u>, without any change in the kinetic energy of the charge.	
3(b)(i)	 	B1 B1
3(b)(ii)	The electric field strength (*E*) at a point between X and Y can be found finding the <u>negative of the gradient of the tangent drawn at that point</u> on the graph in 3(b)(i).	B1
	The electric force acting on the known charge q can be found by <u>multiplying the electric strength *E* at that point with the known charge *q*</u>.	B1
3(c)(i)	Zero	

Question	Marks
3(c)(ii)	



	Electric field lines from positive charge to negative charge shown with arrows.	B1
-	Lines do not intersect	B1
-	Line start and end with charge	
-	Line are closer near the centre line	B1
-	Symmetrical in shape	

Question		Marks
3(c)(iii)	To move the positive charge q at X to infinity:	
	Total electric potential at X due to the charges at Z and Y, Vx = 0	B1
	Work done by external agent to move +e from X to infinity, Wx = e.(0-Vx) = e.0 = 0	
	To move – e at Z to infinity:	
	Total electric potential at Z (Vz) is due to the electric charge at Y = $+e/4\pi\varepsilon_o(0.06)$	B1
	Work done by external agent to move -e from Z to infinity, Wz = -e.(0-Vz) $= -e.\ (0 - e/4\pi\varepsilon_o(0.06))$ $= e^2/4\pi\varepsilon_o(0.06)$ $= 3.837 \times 10^{-27}$ J	
	No work is needed to move the last charge at Y to infinity.	
	Hence total work done by external agent to separate the 3 charges completely = $0+3.837 \times 10^{-27} +0$ $= 3.84 \times 10^{-27}$ J	B1
	Work done by electric field to separate the 3 charges totally from one another $= - 3.84 \times 10^{-27}$ J	
	OR	
	Other variation of method	
	Work done by external agent to assemble charges one at a time = - work done by external agent to separate the 3 charges totally from one another = work done by electric field to separate the 3 charges totally from one another.	
4(a)	The increase in the internal energy of a system is the sum of the external	B1